

Taylor Series

MATH 146 – Calculus II

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Example: Applications of Power Series

Example: Evaluate $\int_0^{\frac{1}{2}} \frac{1}{1-x^3} dx$.

Taylor Series

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Suppose $f(x)$ is a function such that $f^{(n)}(a)$ exists for all $n = 0, 1, 2, \dots$. The the **Taylor series** for $f(x)$ centered at a is the power series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

Note: The Taylor series for a function $f(x)$ may not exist, but if it *does* exist, it gives the power series for the function centered at $x = a$.

Famous Examples of Taylor Series

Example: $f(x) = e^x$.

Example: $f(x) = \cos(x)$.

Example: $f(x) = \sin(x)$.

Note: These are among some power series you should memorize!

Famous Examples of Taylor Series

Example: Find the power series representation for $f(x) = x^2 e^{-x}$.

Example: Evaluate $\lim_{x \rightarrow \infty} x \sin\left(\frac{1}{x}\right)$.

Famous Examples of Taylor Series

Example: Evaluate $\int_0^4 e^{-x^2} dx$.

Famous Examples of Taylor Series

Example: Find the solution to the initial value problem $y' - 4y = 0$, $y(0) = 1$.